Auteur: Abdellah El Moussaoui
Studierichting: Informatica
Oefeningenreeks 7A nr. 10.6

$$\frac{\partial f}{\partial x} = \frac{\frac{\partial (xy)}{\partial x}(x^2+y^2) - (xy)\frac{\partial (x^2+y^2)}{\partial x}}{(x^2+y^2)^2}$$
$$\Leftrightarrow \frac{\partial f}{\partial x} = \frac{y(x^2+y^2) - (xy)(2x)}{(x^2+y^2)^2}$$
$$\Leftrightarrow \frac{\partial f}{\partial x} = \frac{y^3 - yx^2}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{\frac{\partial (xy)}{\partial y}(x^2+y^2) - (xy)\frac{\partial (x^2+y^2)}{\partial y}}{(x^2+y^2)^2}$$
$$\Leftrightarrow \frac{\partial f}{\partial y} = \frac{x(x^2+y^2) - (xy)(2y)}{(x^2+y^2)^2}$$
$$\Leftrightarrow \frac{\partial f}{\partial y} = \frac{x^3 - xy^2}{(x^2+y^2)^2}$$

References